

The Composite Effect of Stock Markets, Bank Stability and Country-Level Institutional Factors on Leverage

ANISHA (BARBORA) NYATEE*

University of Rochester

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Abstract

This paper analyzes the relation between firms' leverage, financial development and firm-level characteristics. I find financial development, measured as stock market turnover ratio or stock market capitalization, is negatively related to firm leverage while financial development, measured as the stability of a country's commercial banking system, is positively related to firm leverage. As stock markets become more developed, leverage decreases. This decrease is more pronounced for smaller firms as compared to larger firms. As banks become more stable, leverage increases and this increase is more pronounced for larger firms as compared to smaller firms. I also find that leverage is lower for firms in politically unstable and more corrupt countries. Conversely, firms are more levered in countries with stronger protection of creditor rights. A higher restrictiveness on foreign direct investment significantly increases leverage ratios of firms.

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1 Introduction

The relation between financial development and economic growth, measured as the log of GDP, is well documented in the literature; however, the effect of financial development on capital structure is still ambiguous. Functioning financial systems ease external financing constraints which impede firm and industry expansion. This paper investigates leverage ratios of firms as a transmission mechanism between financial development, firm-level characteristics based on firm size within each country and capital structure choices across 13 global economies. Intuitively, institutional differences among countries should be an important factor in determining how firms choose to finance their investments. To this end I investigate how firms' leverage ratios differ across countries by accounting for firm characteristics and country-level political, legal and developmental characteristics. In particular, I investigate the simultaneous effect of stock market development and a stable commercial banking system on firms' leverage.

I analyze data from COMPUSTAT North America, COMPUSTAT Global and The World Bank. Although large listed companies have easier access to international financial markets and their financing decisions may be less subject to the institutional constraints imposed by domestic markets, there exist significant differences in their leverage across countries. I show that these differences are associated with the development of financial institutions as well as differences in firm size and characteristics. This paper analyzes the simultaneous effect of the level of stock market development and the stability of the commercial banking system on leverage. I proxy for the level of stock market development in two ways: stock market turnover ratio proxies for the efficiency of the stock market while stock market capitalization as a percentage of GDP proxies for the size of stock market. I use bank z -score as a proxy for the stability of a country's commercial banking system. The stock market turnover ratio and stock market capitalization are a fairly good approx-

imation of the level of development of a country's stock market while the bank z -score is a fairly good approximation of a country's commercial banking system's resilience to stress.

I analyze data from 13 global economies. The results suggest firms operating in countries with a higher stock market turnover ratio tend to have lower leverage ratios. The same holds for firms operating in countries with a higher stock market capitalization as a percentage of GDP. In particular, the interquartile difference in leverage due to stock market development is approximately a -4.9% . This implies that an increase in stock market development explains approximately a 20% decrease in leverage relative to the mean leverage for the full sample. At the same time, I find firms operating in countries with a more stable commercial banking system have higher leverage ratios. The interquartile difference in leverage due to stability of the commercial banking system is approximately 1.1%. Therefore, an increase in the stability of the commercial banking system explains approximately a 4.4% increase in leverage relative to the mean leverage for the full sample. My analyses suggest that the relation between financial development and corporate capital structure is not as intuitive as it may seem. In a broad sense, it seems almost obvious that firms in countries with less developed stock markets will rely more heavily on debt to finance investment and that firms in countries with an unstable commercial banking system will rely more heavily on equity. Although these individual relations between stock market development and leverage and the level of development of the debt market and leverage are well documented in the literature, the relation between financial development and leverage is not obvious since countries with a less developed stock market also have a less developed debt market.

This paper belongs to the literature documenting the importance of legal rules and investor protection rights as important determinants of firms' capital structure choices (La Porta et al., 1997, 1998). I first focus on country-specific financial development determinants of firm leverage ratios. I build on previous literature by using firm-level data as well

as country-level development proxies as opposed to using aggregate firm data for cross-country comparisons (Mayer ,1988; Demirgüç-Kunt and Maksimovic, 1998; Booth et al., 2001). I employ the Ordinary Least Squares (OLS) technique to identify a causal relation between country-level stock market development and stability of the commercial banking system and firm-level capital structure. Although a number of papers have attempted to understand the relation between stock market development and firms' financing decisions, few have studied the simultaneous effect of both the stock market and the banking system on firms' leverage. Broadly, the literature documents economies with developing stock markets seem to dominate the higher end of the leverage spectrum while economies with developed stock markets dominate the lower end of the leverage spectrum. I find this is not necessarily the case as economies with developed stock markets are also the economies with a developed banking system which makes borrowing more lucrative than issuing equity.

Giannetti (2003) limits her sample to private firms in order to reduce the possibility of spurious correlations as well as bias related to firm size. If large companies bias the results, then the leverage across countries should not be significantly affected by the level of development. However, this is not supported by the data. I extend Giannetti's work in that I analyze data from a sample of large listed companies. I control for firm size by estimating quartile regressions where I divide the data into four quartiles based on firm size within each country in my sample. The regression estimates across quartiles are qualitatively similar to the full sample estimates. I also employ fixed effects in the regressions. The firm fixed effects control for unobserved, time invariant heterogeneity across firms and therefore reduce the possibility of spurious correlations.

In studying the simultaneous effect of the development of stock markets and banking system on leverage, I find that an active stock market, measured as the stock market turnover ratio is a first order determinant of firms financing choices. The literature focusses on the country-level effect by aggregating the firm-level characteristics at the country-level.

In my analyses, I retain the firm-level data. I then look at the effect of firm-level characteristics together with country-level political, legal and developmental controls on leverage.

The rest of the paper is organized as follows. Section 2 discusses the literature. Section 3 describes the data and develops relevant hypotheses. Section 4 outlines the empirical design and methods. Section 5 discusses the results. Section 6 describes alternate tests and robustness checks. Section 7 concludes.

2 Literature Review

Prior literature in empirical corporate finance shows corporate capital structure decisions depend on various firm attributes such as availability of collateral (Titman and Wessels, 1988; Barclay and Smith, 1995) which proxy for agency problems and asymmetric information. However, these studies focus primarily on U.S. companies and therefore fail to capture the effect of institutional differences on capital structure decisions. There is also a wide literature on cross-country comparisons of capital structure that use aggregate data on firm and country characteristics (Mayer, 1988). These studies fail to account for the influence of firm characteristics on capital structure decisions in different institutional environments. A notable exception is Rajan and Zingales (1995), who use balance sheet data from listed companies in G7 economies to show factors identified previously as determinants of firm leverage in the U.S. are similarly correlated in these countries. This is intuitive, given G7 countries are the world's seven most industrialized economies. In a related paper, Booth et al. (2001) use aggregate firm data to show debt ratios in developing countries are generally affected in the same way and by the same characteristics as developed countries. Although the signs on the explanatory variables are mostly as expected, they differ across countries and the overall impact of their model is low.

I use firm fixed-effects in the leverage equations to study the interaction between firm-

and country-level characteristics and their impact on leverage for firms operating in different institutional environments. In particular, I analyze the simultaneous effect of stock market development and bank stability together with firm-level controls that have been identified in the corporate finance literature as determinants of firms' financial structures and other country-level political and legal controls. Demirgüç-Kunt et al. (2012) show financial systems become more market oriented as economies develop. In particular, they show banks and securities markets increase in size relative to the economy. I follow Demirgüç-Kunt et al. (2012) in studying the importance of banks and securities markets around the world, and extend this literature by analyzing the simultaneous effect of the level of development of banks and securities markets on firms' leverage across countries. Intuitively, firms operating in countries with less developed stock markets experience higher leverage while those in countries with an unstable commercial banking system have lower leverage ratios. These two aspects of financial development have opposing effects on leverage thereby making the overall impact of financial development on leverage more complicated than what it may seem.

There is also an extensive literature examining the relation between developmental economics and financial development. Prior work in this area argues better functioning financial systems ease the external financing constraints that impede firm and industrial expansion. Demirgüç-Kunt and Levine (2008) show companies in countries with better developed financial systems have better access to external financing. Much of the developmental economics and financial development literature uses the log of GDP as a proxy for economic growth and studies the effect of financial sector development on GDP, attributing a causal relation between financial development and economic growth. I focus my work on studying the effect of financial development on firms' financing choices. To this end, I use firms' leverage as a transmission mechanism to study the relation between financial development and capital structure decisions.

Giannetti (2003) examines how firm characteristics, legal rules and financial development affect corporate finance decisions. She uses data on unlisted companies in 26 European countries to show institutions play an important role in determining the extent of agency problems. Giannetti documents the fact that firms are more levered in countries with less developed stock markets. She further claims large listed companies have easier access to international financial markets and hence their corporate finance decisions are less subject to institutional constraints imposed by domestic markets. Building on this, I show institutional constraints together with firm-level characteristics are in fact an important determinant of corporate capital structure decisions, even for large listed companies.

Demirgüç-Kunt and Maksimovic (1995) primarily look at the effect of stock market development on firm's financing choices. They control for the importance of financial intermediaries using three empirical indicators. I extend this work by not only analyzing the effect of stock market development on leverage but also controlling for the stability of the commercial banking sector and other country-specific legal and political controls.

Fan et al. (2002) examine capital structure and debt maturity choices of firms in a cross-section of 47 developed and developing countries. They focus on a country's legal system, public governance, taxation and market participation. Their analysis is at the country level. They document that countries with a developing stock market dominate the higher leverage range while countries with a well-developed stock market tend to be at the lowest range. I find that this is not true in the data. Firms in countries with the most developed stock markets have some of the highest leverage ratios while firms in countries with lesser developed stock markets have lower levels of leverage.

Therefore, I argue that the relation between the level of development of stock markets and leverage is not as simple as the literature suggests. The banking and securities markets increase in size relative to the economy. Hence countries with a well-developed stock market also have a well-developed banking sector. This results in a more competitive debt

market where interest rates are low and hence firms' find it more lucrative to increase their debt position, thereby increasing leverage.

I extend literature by focusing on firm-level data and carrying out my analysis across the full sample as well as various sample splits to identify the combined effect of the level of development of the stock market as well as the stability of the banking system on firm's financing decisions. I also include more proxies for institutional differences, emphasizing the role of political stability and the strength of legal rights as well as foreign direct investment regulations in firms' financing choices.

3 Discussion of Data and Hypotheses Development

I limit my analysis to a group of 13 global economies in which data for a sufficient number of firms are available. My sample consists of four high income OECD economies (Australia, Canada, New Zealand and the United States), six upper-middle income economies (Argentina, Brazil, Mexico, Malaysia, Peru and South Africa) and three lower-middle income economies (Egypt, Indonesia and India), as classified by the World Bank. The final sample consists of countries with data on 20 or more firms spanning at least five years. I analyze a mix of different income level economies and require sufficient data in creating the final sample so I can compare leverage ratios across different levels of institutional development as well as to make sure that the sample is representative of the country in question.

The firm-level balance sheet data come from Standard and Poor's COMPUSTAT Global and COMPUSTAT North America databases. The firm characteristics include leverage ratio, firm size, investments, accounting standards, tangible assets and profitability. Several alternative definitions of leverage are used in prior literature. The primary difference is whether book or market measures are used. As noted by Frank and Goyal (2009), book ratios are conceptually different from market ratios. Market values are forward looking

while book values are backward looking. Since my study attempts to document leverage ratios, and hence firm financing decisions based on relative claims of firm value using debt and the level of development of financial institutions, I focus on leverage measures based on book debt values.

Some of the leverage ratio measures also differ based on whether all debt or only long-term debt is considered. The primary specification of my model analyzes the impact of firm- and country-level factors on leverage defined as the ratio of total debt to book assets. I calculate total debt as the sum of long-term debt and debt in current liabilities. Booth et al. (2001) use leverage defined as the ratio of total liabilities to book assets, as well as the ratio of total liabilities less current liabilities to book assets less current liabilities. A firm's total liabilities include liabilities that are not limited to long-term debt and debt in current assets. Since I look at how leverage and therefore debt issuance differs across countries, a measure of leverage as defined by Booth et al. (2001) is not very informative for the purpose of my analysis.

Firm size is calculated as the log of book assets. Larger firms tend to be more diversified and tend to fail less often. Interpreting size to be an inverse proxy for the probability for bankruptcy, I anticipate firm size to have a positive influence on the supply of debt. Therefore larger firms should be more levered. *Investments* is calculated as the sum of capital expenditures and increase in investments and acquisitions less sale of property, plant and equipment. Interpreting *Investments* as a proxy for growth opportunity, firms expecting high future growth would rather raise new capital for investment using debt over equity. This is because profits are associated with growth opportunities and had the firm issued equity, this profit would have to be shared with the buyers of the new stocks. Thus, I anticipate increased investment to be associated with increased leverage.

The accounting standard measure, *IFRS*, is an indicator variable for firms that follow domestic accounting standards generally in accordance with or fully compliant with the

International Financial Reporting Standards (IFRS). 10,159 firms in the sample are IFRS compliant. To distinguish between the IFRS and the U.S. GAAP compliant firms, I include another indicator variable, *GAAP*. This variable takes a value of one if the firm operates in the U.S., follows U.S. accounting standards, or follows domestic standards generally in accordance with U.S. GAAP. 39,972 firms in the full sample are U.S. GAAP compliant. I include these variables in the regression specification as proxies for information asymmetry. The IFRS and GAAP outline a set of principles that companies are required to follow in preparing their financial statements. These statements are a critical source of information for the company's shareholders and debtors as they aid in understanding the company's financial performance and the handling of the company's resources. Firms that are IFRS and GAAP compliant are more transparent in their financial reporting thereby making equity a preferred choice for financing investment; therefore, I anticipate a negative relation between leverage and the *IFRS* and *GAAP* variables.

I calculate *Tangibility* as the ratio of total property, plant and equipment to book assets. Assets serve as collateral thereby diminishing the risk of the lender suffering the agency costs of debt. This implies that the greater the proportion of tangible assets the more willing the lenders should be to issue debt. Therefore, the relation between tangibility and leverage should be positive. *Profitability* is calculated as the ratio of EBITDA to total book assets. The relation between profitability and leverage is ambiguous. Myers and Majluf (1984) predict a negative relation because more profitable firms will prefer to finance investment with internal funds rather than debt. On the other hand, Jensen (1986) predicts a positive relation between profitability and leverage conditional on the fact that the market for corporate control is effective. In the case of an ineffective market for corporate control, managers prefer to avoid the disciplinary role of debt which would lead to a negative relation between profitability and debt.

The final sample consists of firms for which all of these data are available. For some

firms, COMPUSTAT lists multiple data points per year. Since the entries for each variable are not identical, I interpret this as a data error and exclude these firms from the final sample. I also exclude firms that are listed on multiple exchanges from the data set. The final sample is an unbalanced panel of 106,182 observations, ranging from 2002 to 2011.

Data on the development of financial institutions are from the World Bank's Global Financial Development Database (GFDD). I look primarily at three different dimensions of financial development. *Stock Market Turnover Ratio* is defined as the total value of shares traded during a given period divided by the average market capitalization for that period. Average market capitalization is calculated as the average of the end-of-period values for the current period and the previous period. This is essentially a liquidity measure. Liquid capital markets are markets where it is relatively inexpensive to trade financial instruments and where there is relatively little uncertainty regarding the timing and settlement of those trades. Therefore, liquid markets improve the allocation of capital and enhance prospects for long term economic growth. Thus, for firms operating in countries with a greater stock market turn-over ratio, the leverage ratio should be smaller. *Stock Market Capitalization* is calculated as the share price times the number of shares outstanding. Market size is positively correlated with the ability to mobilize capital and diversify risk. Therefore I anticipate a negative relation between leverage and stock market capitalization as a percentage of GDP. Finally, bank *z*-score explicitly compares capitalization and returns with the volatility of returns to measure a bank's solvency risk.¹ Therefore, a higher *z*-score implies a lower probability of insolvency and hence a more stable commercial banking sys-

¹The *z*-score is calculated as follows:

$$z = \frac{k + \mu}{\sigma}, \text{ where}$$

k = equity capital as a percent of assets
 μ = return as a percent of assets
 σ = standard deviation of return on assets

tem. Therefore for firms operating in countries with a higher bank z -score, leverage should be higher.²

Data for country-level government controls such as *Political Stability* and *Control of Corruption* are taken from the World Bank's Worldwide Governance Indicators database (WGI). These data are gathered from a number of survey institutes, think tanks, non-governmental organizations, international organizations, and private and public sector firms. Estimates for these measures give the country's score on the aggregate indicator, in units of a standard normal distribution ranging from -2.5 to 2.5. The WGI data are available by country for the period between 2002 and 2011 which limits the final sample to this time period.

The WGI defines *Political Stability* to capture perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. For firms operating in politically unstable economies, structural change is more likely and this creates uncertainty about the post-change regime. Investors are uncertain about the future policy choice as the new government decides which policy to adopt (Pástor and Veronesi, 2013). Under political uncertainty stock prices are more volatile and more highly correlated (Bittlingmayer, 1998; Voth, 2002; Santa-Clara and Valkanov, 2003) making equity more sensitive to political uncertainty. It then follows that if firms in politically unstable economies raise finance through equity issuance they are forced to issue equity at a lower price to afford sufficient risk premium to provide protection against the risk associated with political instability. These firms would then choose to finance investment via debt as this would be less mispriced compared to equity. Intuitively, firms in more politically unstable economies should have higher leverage ratios.

The measure for *Control of Corruption* captures the strength and the effectiveness of a country's policy and institutional framework to prevent and combat corruption. Thus

²One limitation of this measure of stability is that the z -score is based purely on accounting data and hence the measure is only as good as the underlying accounting framework.

a lower number for this index implies a higher rate of corruption. It is tougher for firms operating in more corrupt countries to seek external financing via debt due to corruption in the bank supervisory policies.

Data on the *Strength of Legal Rights*, taken from the World Bank's Global Financial Development Database, measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders thereby facilitating lending. The index ranges from zero to 10, with higher scores indicating that these laws are better designed to expand access to credit. The strength of legal rights data are available for the period of 2004-2011 and is constant over time for nine of the 13 countries. Hence, I fit a time series model to this data and extrapolate from this model the index measure for years 2002 and 2003. The protection investors receive will determine their readiness to finance firms' investments. It is intuitive, then, that for firms in countries with stronger protection of creditor rights debt issuance is higher than for firms in countries with weaker protection of creditor rights.

The Foreign Direct Investment Restrictiveness Index (*FDIRR*) measures statutory restrictions on foreign direct investment. This index gauges the restrictiveness of a country's Foreign Direct Investment (FDI) policy by looking at foreign equity limitations, restrictions on the employment of foreigners, screening or approval mechanisms and restrictions on branching and on capital repatriation. Smaller values for *FDIRR* imply fewer restrictions on foreign direct investment. More restrictions on foreign direct investment imply that firms are more reliant on either their domestic stock market for equity or on domestic creditors for debt. The relation between leverage and *FDIRR* is ambiguous. Countries with greater restrictions on foreign direct investment are also countries with a less developed stock market and a more unstable commercial banking system. I include this variable in the regression specification to control for the possibility of firms raising capital in foreign markets through foreign investors and foreign subsidiaries. This data is available on the OECD website.

Debt is tax deductible while equity is not. Hence for firms in countries with a high corporate tax rate, debt should be a preferred choice for financing investment. Given this logic I also include the *Corporate Tax Rate* for the countries in my sample in the regression specification. The corporate tax rate data is made available by KPMG International through their Corporate and Indirect Tax Survey.

Finally, financial and utility firms are excluded from the sample. The summary statistics for the firm-level controls for the full sample are given in Table 1. The mean leverage for the full sample is approximately 0.243. The average firm size for the full sample is approximately 1.70 taken as log of total assets, where total assets is in millions of U.S. dollars; while average annual investment for the full sample is approximately 61.50 million U.S. dollars. The summary statistics for the firm-level controls by country of incorporation are listed in Table 2. The summary statistics for the country-level controls are listed in Table 3.

4 Methods

COMPUSTAT Global reports data in the local currency. Therefore, I use foreign exchange rate data made available by the Federal Reserve Board to convert all data to U.S. dollars by averaging daily exchange rates on an annual basis. The firm-level data from COMPUSTAT and the country-level data from the World Bank, OECD and KPMG International are merged on the basis of country and year. Due to the existence of outliers in the firm-level characteristics the data are winsorized at the 1% and 99% levels for firms in each country. A modest goal of this paper is to study the comparative capital structure decisions where the units are firms in a country. For each firm I observe annual data on a variety of firm-specific variables as well as country-specific government, legal and development controls. I account for firm fixed effects by demeaning the variables at the firm level. To control

for year fixed effects I include a dummy variable for each year in the sample. Since the final sample does not include firms listed on multiple exchanges I do not include dummy variables for the countries in the sample. The firm and year fixed effects will absorb any country invariant heterogeneity. I first estimate an OLS regression model of the following form:

$$\begin{aligned}
Leverage_{i,j,t} = & \beta_0 + \beta_1(Financial\ Development\ Proxy)_{j,t} + \beta_2(Firm\ Size)_{i,t} \\
& + \beta_3(Investment)_{i,t} + \beta_4(IFRS)_{i,t} + \beta_5(GAAP)_{i,t} \\
& + \beta_6(Tangible\ Assets)_{i,t} + \beta_7(Profitability)_{i,t} \\
& + \beta_8(Political\ Stability)_{j,t} + \beta_9(Strength\ of\ Legal\ Rights)_{j,t} \\
& + \beta_{10}(Control\ of\ Corruption)_{j,t} + \beta_{11}(FDIRR)_{j,t} \\
& + \beta_{12}(Corporate\ Tax\ Rate) + \gamma(Year\ Dummies) + \epsilon_{i,j,t}
\end{aligned}$$

Here, $Leverage_{i,j,t}$ represents the leverage of firm i in country j at time t . In this equation the *Financial Development Proxy* variable is specified to be either the *Stock Market Turnover Ratio*, the *Stock Market Capitalization* as a percentage of GDP or the *Bank z-score*. The regression estimates suggest that the coefficient on the *Financial Development Proxy* variable is statistically insignificant for all three specifications of this variable. This seems counterintuitive as institutional differences among countries should be an important factor in determining how firms choose to finance their investments. In each of the three specifications of the *Financial Development Proxy* variable I only account for either the development of the stock market or the stability of the commercial banking system each time. However, one cannot ignore the fact that both banking systems and stock markets simultaneously affect firms' financing decisions. In countries with lesser developed stock markets, the banking system is also more unstable thereby making it harder for firms to seek loans from banks. Therefore, controlling for stock market development and ignoring

the stability of the banking system and vice versa biases the OLS results.

To control for the simultaneous effect of stock market development and stability of the commercial banking system on firms' leverage, I estimate two separate OLS regressions, one including the *Stock Market Turnover Ratio* and the *Bank z - score* as explanatory variables and another including the *Stock Market Capitalization* and the *Bank z - score* as explanatory variables:

$$\begin{aligned} Leverage_{i,j,t} = & \alpha_0 + \alpha_1(Stock\ Market\ Development\ Proxy)_{j,t} + \alpha_2(Bank\ z - score)_{j,t} \\ & + \alpha_3(Firm\ Size)_{i,t} + \alpha_4(Investment)_{i,t} + \alpha_5(IFRS)_{i,t} + \alpha_6(GAAP)_{i,t} \\ & + \alpha_7(Tangible\ Assets)_{i,t} + \alpha_8(Profitability)_{i,t} + \alpha_9(Political\ Stability)_{j,t} \\ & + \alpha_{10}(Strength\ of\ Legal\ Rights)_{j,t} + \alpha_{11}(Control\ of\ Corruption)_{j,t} \\ & + \alpha_{12}(FDIRR)_{j,t} + \alpha_{13}(Corporate\ Tax\ Rate) + \delta(Year\ Dummies) + \nu_{i,j,t} \end{aligned}$$

Here, *Stock Market Development Proxy*_{*j,t*} is specified to be either the stock market turnover ratio or the stock market capitalization of country *j* at time *t*. Controlling for the stock market turnover ratio or stock market capitalization and the bank *z*-score simultaneously results in the statistically significant coefficients for both these variables. Holding all firm and country-level controls constant, there is a significant negative correlation between leverage and both specifications of the *Stock Market Development Proxy* variable. Ceteris paribus, there is a significant positive relation between leverage and *Bank z - score*. The *F*-statistic for joint significance of *Stock Market Turnover Ratio* and *Bank z - score* is 18.23 while that for the joint significance of *Stock Market Capitalization* and *Bank z - score* is 25.87. The standard errors are clustered by firm to control for the possibility that residuals of the estimates are correlated across firms.

Since all the data in my sample are of large listed companies, one concern is that firm size as opposed to stock market development and bank stability drives the main result.

This is plausible since it is relatively easier for larger companies to access international financial markets where they can raise debt or equity through their foreign subsidiaries; hence their corporate finance decisions are less subject to the institutional constraints imposed by domestic markets (Giannetti, 2003). It is therefore possible that the presented model specification fails to account for this subsidiary effect that may likely exist for larger listed companies. To tackle this problem I divide the data for each country into quartiles based on firm size within each country and then estimate the previously specified OLS regression for each quartile. The estimates for each quartile are qualitatively similar to those for the full sample. The firm-level summary statistics by quartile are listed in Table 4.

5 Results

As a first step in understanding the relation between financial development and capital structure by analyzing the effect of stock market turnover ratio, stock market capitalization and bank z -score on firm leverage, I estimate various specifications of the OLS regression model outlined above for the full sample. The regression estimates are presented in Table 5. The coefficient on the *Financial Development Proxy* variable is statistically insignificant in the first three specifications of the OLS regression where I control for either *Stock Market Turnover Ratio*, *Stock Market Capitalization* or *Bank z – score*. This seems counterintuitive as common sense would suggest that firms in countries with less developed stock markets will rely more heavily on debt to finance investment and that firms in countries with an unstable commercial banking system will rely more heavily on equity. Controlling for stock market development and ignoring the stability of the banking system and vice versa biases the OLS results. Therefore, as a next step I estimate two additional specifications, (4) and (5), where I control for the level of

development of the stock market and the commercial banking system simultaneously. *Stock Market Turnover Ratio* and *Stock Market Capitalization* as a percentage of GDP proxy for the level of development of the stock market while *Bank z - score* is a proxy for the stability of the commercial banking system. This results in statistically significant coefficients on the *Stock Market Turnover Ratio* (and *Stock Market Capitalization*) and *Bank z - score*. The regression estimates for the full sample suggest for a 1% increase in *Stock Market Turnover Ratio*, firms' *Leverage* decreases by 0.1% holding all other firm- and country-level controls constant. In terms of economic significance of the estimates this translates to the fact that in going from the twenty fifth to the seventy fifth percentile of stock market turnover ratio, there is approximately a 4.9% decrease in leverage. The mean leverage for the whole sample is approximately 25%. Therefore, an increase in stock market turnover ratio explains approximately a 20% decrease in leverage relative to the mean. The results are identical for *Stock Market Capitalization*.

Holding all other firm- and country-level controls constant, for a one unit increase in the *Bank z - score*, firms' leverage increases by 0.2%. Intuitively, as the activity of stock markets or the size of stock markets increase relative to the stability of the commercial banking system, leverage decreases as firms are able to increase their equity position more easily. The reverse is true when the commercial banking system becomes more stable relative to the size and activity of the stock market. However, in terms of the economic significance of the influence of the commercial banking system on firms' leverage, the evidence is not as strong. In going from the twenty fifth to the seventy fifth percentile of bank z -score, there is approximately a 1.1% increase in leverage. Relative to the mean leverage for the sample this only explains approximately 4.4% of the increase in leverage.

In contrast to Demirgüç-Kunt and Maksimovic (1995) I find a significant negative relation between *Stock Market Turnover Ratio* and *Leverage*. This suggests that an active stock market is a first order determinant of firms' financing choice. I also document

that the negative linear relation between *Stock Market Turnover Ratio* and *Leverage* and *Stock Market Capitalization* and *Leverage* and the positive linear relation between *Bank z-score* and *Leverage* remains statistically significant for the full sample even in the presence of firm-level controls that have been identified in the corporate finance literature as determinants of firms' financial structure. Demirgüç-Kunt and Maksimovic find these relations to hold only in various sub-samples of their analysis.

Tables 6 and 7 present the estimates of the quartile regressions when the *Stock Market Development Proxy* variable is specified to be the *Stock Market Turnover Ratio* and *Stock Market Capitalization* respectively. The OLS regression estimates for the quartile regressions are qualitatively similar to those of the full sample. The regression results in Tables 6 and 7 allow for a better understanding of the simultaneous effect of stock market development and bank stability on leverage. Looking at the regression estimates for quartiles 1 and 2, as security markets become more developed smaller firms are more likely to move away from debt financing towards equity financing. As stock markets develop, a larger number of smaller firms have easier access to the stock markets in terms of making equity offerings. Small firms are able to increase their equity position more easily by means of seasoned equity offerings thereby reducing leverage. Therefore, holding bank *z*-score constant, as stock markets develop, the leverage of small firms decrease significantly compared to that of larger firms. On the other hand, the regression estimates of the third and fourth quartile provide suggestive evidence that as banks become more stable, larger firms are more likely to move away from equity financing towards debt financing. With a more stable commercial banking system in place larger firms find debt financing more lucrative as compared to issuing equity as the debt market becomes more competitive thereby reducing interest rates. Larger firms are thus able to increase leverage by issuing debt. Therefore, holding stock market development constant, as banks become more stable, the leverage of larger firms increase significantly compared to that of smaller

firms. The economic significance of stock market development measured as stock market turnover ratio and stock market capitalization is decreasing across quartiles based on firm size while that of bank stability is increasing across quartiles based on firm size. However, the economic significance across quartiles remains qualitatively similar to the estimates for the full sample.

A difference of means test suggests that stock market development is indeed more beneficial to smaller firms as compared to larger firms. Although the quartile regressions provide some suggestive evidence that bank stability is likely to benefit larger firms more than smaller firms the difference in means test does not capture this effect. This is possibly due to the fact that the bank z -score variable fails to capture accurately the debt that firms issue. To be able to study in depth the relation between bank development and leverage it becomes imminent to account for the different kinds of debt that the firm issues. This is left for future work.

Consistent with Rajan and Zingales (1995) and Booth et. al (2001) I find that leverage ratios for firms in developing countries are affected by the same variables as firms in developed countries. Among firm-level characteristics, firm size, tangibility and profitability have a statistically significant impact on leverage. As mentioned earlier, if we think of firm size as an inverse proxy for bankruptcy, the effect of firm size on equilibrium leverage should be positive irrespective of how leverage is defined. This is true in the data. Larger firms are more levered as compared to smaller firms. Tangible assets serve as collateral for debt thereby allowing firms to lever up by issuing debt to raise funds for the purpose of investment. The data supports this view. Firms with a larger fraction of tangible assets are more levered as compared to firms with a smaller fraction of tangible assets.

As far as the impact of profitability on leverage is concerned, the regression estimates suggest a negative relation. This is in agreement with Myers and Majluf (1984) where they show that more profitable firms prefer to finance investment with internal funds rather

than debt. This results in more profitable firms having lower leverage as compared to less profitable firms. It is worth noting that this result does not sit well with the static trade-off theory which predicts that highly profitable firms issue more debt in order to lower their tax bill.

The effect of investment on leverage is neither statistically nor economically significant in either of the regression specifications. According to the regression estimates, although the sign on the *IFRS* variable is as anticipated, it does not have a significant effect on firms' leverage. However, U.S. GAAP compliant firms have higher leverage as compared to firms that are not compliant with U.S. GAAP. Firms that are more transparent in their financial reporting are levered up compared to their less transparent counterparts.

Among country-level controls, there is significant evidence in favor of the fact that firms in politically stable economies are less levered than those in politically unstable economies. As discussed earlier, under political uncertainty stock prices are more volatile and highly correlated, making equity more sensitive to this uncertainty. Therefore it is beneficial for firms to finance investment via debt, as debt will be less mispriced compared to equity. There is evidence in the data that, for firms in countries with more secured creditor rights, leverage is higher than that of firms in countries where creditor rights are not as strongly protected. For firms in countries with a higher rate of corruption, leverage is lower than that of firms in countries with an effective policy and institutional framework to prevent and combat corruption.

As per the regression results, there exists a positive relation between *FDIRR* and leverage. This is interesting as the countries with greater restrictions on FDI are also the countries with a more unstable commercial banking system. To completely understand the relation between FDI regulations and leverage would require a deeper understanding of the debt market. This is left for future work.

Finally, the sign of the coefficient on *Corporate Tax Rate* is as expected. The tax code

for countries allows corporations to deduct interest payments as an expense but dividend payments to stock holders are not deductible. This differential treatment encourages firms to issue debt over equity. Therefore firms in countries with a high tax rate are more levered.

6 Robustness

As can be seen from the country-level summary statistics in Table 3 the stock market turnover ratio, stock market capitalization and bank z -score measures have very large variation for some countries. This helps illustrate the fact that these measures are possibly noisy proxies for the effects I am trying to study. To ensure that it is not the noise that is driving the relation between the level of development of the stock market and leverage and bank stability and leverage I use a time series of fitted values for these measures. In unreported regressions I estimate the previously specified OLS regression including time-series fitted values for both the proxies for stock market development as well as bank z -score together with other firm- and country-level controls. The results remain qualitatively unchanged for the full sample as well as the quartile regressions.

Although most studies focus on a single measure of leverage, it is important to confirm crucial results are robust to alternate measures of leverage. Hence, as a check for robustness I estimate my model using the ratio of long-term debt to assets as a proxy for leverage. Table 8 presents the OLS estimates for the full sample. The results are qualitatively similar to the main regression estimates presented in specifications (4) and (5) of Table 5. I also estimate quartile regressions for this measure of leverage. Tables 9 and 10 report the results of the OLS regressions for each quartile. Again, the results are qualitatively similar to those presented in Tables 6 and 7 respectively.

In unreported regressions I employ alternate measures of tangibility and profitability. I calculate tangibility as the ratio of the difference of book assets and current assets to

book assets. Profitability is calculated as the ratio of EBIT to book assets. I also use an alternate measure for the strength of legal rights. The legal rights variable is as defined by La Porta et al (1997). This variable is an index aggregating different creditor rights. The index ranges from zero to four and is formed by adding a value of one for each of the following criteria: 1) the country imposes restrictions to file for reorganization; 2) secured creditors are able to gain possession of their security once the reorganization petition has been approved; 3) secured creditors are ranked first in the distribution of the proceeds that result from the disposition of the assets of a bankrupt firm; and 4) the debtor does not retain the administration of its property pending the resolution of the reorganization. The main results of my analysis remain qualitatively unchanged for these alternate specifications of the variables in question.

Hart (1995) shows that the defining feature of various securities is the rights they bring to their owners. Shareholders typically possess the right to vote for directors of companies while bondholders possess the right to repossession collateral when companies fail to make promised payments. La Porta et al. (1998) show this view of securities being inherently characterized by some intrinsic rights to be incomplete. They argue that these rights depend on the legal rules of the jurisdictions in which they operate. Further, they show that countries with a legal system that originated from the British common law offer creditors stronger legal protections against managers compared to those with a legal system that originated from the French, German or Scandinavian civil law. Therefore, in an alternate specification of creditor rights I include an indicator variable for countries whose legal system originated from the French, German or Scandinavian civil law systems. Since the legal system is more favorable to investors in common law countries, leverage should be lower for firms in countries whose legal rules originated from the French, German or Scandinavian civil law systems as compared to countries whose legal rules originated from the British common law system. Hence, the results are qualitatively similar to the

main regression specification where leverage is higher for firms in countries with stronger protection of creditor rights (common law countries).

7 Conclusion

This paper documents that leverage ratios are higher for firms operating in countries with less developed stock markets while leverage ratios are lower for countries with an unstable commercial banking system. I use firm-level data on listed companies in 13 global economies and show that the level of development of financial institutions together with firm characteristics and country-level political, legal and developmental controls plays an important role in capital structure decisions. Using the OLS approach I show that for firms operating in countries with a higher stock market turnover ratio, the leverage ratios are significantly lower than those operating in countries with a lower stock market turnover ratio. The results are qualitatively similar for firms in countries with a higher stock market capitalization as a percentage of GDP as compared to firms in countries with a lower stock market capitalization. At higher levels of financial development domestic stock markets tend to become more active and efficient relative to domestic banks while at low levels of financial development commercial banks tend to dominate the financial system, making debt issuance the viable choice for financing investment, thereby increasing leverage. The estimates also suggest that for firms in countries with a more stable commercial banking system, leverage is on average significantly higher than for firms in countries with an unstable commercial banking system.

Countries with more developed stock markets tend to rely less on debt for the purposes of financing investment. However, countries with more developed stock markets are also countries with a more stable commercial banking system which makes debt financing a viable option. Although the negative linear relation between *Stock Market Turnover*

Ratio and *Leverage* and *Stock Market Capitalization* and *Leverage* and the positive linear relation between *Bankz – score* and *Leverage* is consistent across quartiles and the full sample estimates, the quartile regressions highlight an interesting pattern. As stock markets develop, the leverage of small firms decreases significantly compared to that of larger firms as small firms move from debt to equity financing. A difference in means test confirms that stock market development benefits smaller firms more than it does larger firms. Although the quartile regressions provide suggestive evidence that as banks become more stable, the leverage of larger firms increases significantly compared to that of smaller firms the difference in means test does not confirm this result.

In addition, I show that leverage is lower for politically stable countries. With an unstable political system comes an increased probability of structural change, making investors uncertain about the future policy choice. Under political uncertainty stock prices are more volatile and more highly correlated, making equity more sensitive to this uncertainty. Under such circumstances debt is less mispriced and is hence a preferred choice for financing investment thereby increasing leverage. The data also supports the view that, for firms in countries with a better system in place to control and combat corruption, leverage is, on average, higher.

My results are consistent with the prior literature in that I find capital structure decisions in developing countries are affected by the same variables as in developed countries (Rajan and Zingales, 1995). The firm-level characteristics affect leverage in the same way as has been established in the literature (Booth et al., 2001). The protection investors receive will determine their readiness to finance firms' investments. Therefore for firms in countries with stronger protection of creditor rights debt issuance is higher than for firms in countries with weaker protection of creditor rights (La Porta et al., 1997, 1998; Hart, 1995). It is also true in the data that firms are more levered in countries with a less developed stock market (Giannetti, 2003).

To be able to completely understand the interaction of country-level political, legal and developmental factors and how they influence capital structure decisions, much needs to be done to account for the different types of debt that are being issued by firms around the world and for the different factors of financial development that have opposing influences on leverage. Since it is clear that firms issue different kinds of debt, and that they issue debt from creditors other than just commercial banks, in extending this work, I intend to decompose the debt market and study the simultaneous effect of stock market development and debt market development on firms' leverage. I would also like to extend this project by analyzing the effect of the institutional environment in which firms operate together with firm-level characteristics on the types of debt that firms choose to issue. To start, more and reliable data on firms outside of the U.S. and Canada is needed. This study includes only 13 global economies, however, to get a better understanding of capital structure decisions in the developing and the developed world it would be helpful to be able to analyze data on a larger number of countries. Looking at different sample splits based on the overall level of development of a country as well as other firm-level determinants of leverage would possibly give a deeper understanding of capital structure choices of firms operating in different institutional environments.

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Table 1: Firm-level Summary Statistics (Full Sample)

Variable	N	Mean	Standard Deviation
Leverage	106,182	0.243	0.278
Firm Size	106,182	1.699	1.028
Investment	106,182	61.499	442.008
Tangibility	106,182	0.300	0.250
Profitability	106,182	-0.062	0.433

The table shows summary statistics of the firm-level characteristics for the whole sample. *Leverage* is calculated as the ratio of total debt to book assets. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant and equipment. *Tangibility* is calculated as the ratio of total property, plant and equipment to book assets while *Profitability* is defined as the ratio of EBITDA to total book assets.

Table 2: Firm-level Summary Statistics (by Country)

Country	N	Summary Statistic	Total Debt Book Assets	Firm Size	Investment	Tangibility	Profitability
Argentina	499	Mean:	0.196	2.141	89.639	0.382	0.142
		Standard Deviation:	0.183	0.835	280.853	0.217	0.100
Australia	13,338	Mean:	0.130	1.297	49.037	0.320	-0.154
		Standard Deviation:	0.176	0.967	542.904	0.287	0.392
Brazil	2,395	Mean:	0.316	2.463	257.628	0.325	0.099
		Standard Deviation:	0.269	0.947	1,703.328	0.211	0.108
Canada	11,883	Mean:	0.167	1.828	112.047	0.433	-0.100
		Standard Deviation:	0.199	1.086	525.524	0.313	0.365
Egypt	420	Mean:	0.214	2.448	96.974	0.403	0.134
		Standard Deviation:	0.183	0.678	316.998	0.196	0.095
Indonesia	2,591	Mean:	0.304	1.988	42.819	0.393	0.106
		Standard Deviation:	0.243	0.766	169.129	0.226	0.095
India	22,401	Mean:	0.283	1.481	38.739	0.329	0.101
		Standard Deviation:	0.205	0.878	302.143	0.205	0.080
Mexico	922	Mean:	0.222	3.006	410.086	0.440	0.121
		Standard Deviation:	0.163	0.737	1,513.722	0.198	0.068
Malaysia	7,807	Mean:	0.216	1.825	20.321	0.340	0.072
		Standard Deviation:	0.184	0.660	113.339	0.205	0.087
New Zealand	896	Mean:	0.235	1.800	35.524	0.354	0.076
		Standard Deviation:	0.185	0.899	121.028	0.275	0.188
Peru	684	Mean:	0.209	2.180	33.332	0.479	0.147
		Standard Deviation:	0.150	0.621	80.761	0.213	0.102
U.S.A	39,972	Mean:	0.283	1.750	51.217	0.215	-0.185
		Standard Deviation:	0.363	1.128	202.655	0.223	0.595
South Africa	2,374	Mean:	0.169	2.106	94.583	0.273	0.138
		Standard Deviation:	0.149	0.934	321.031	0.225	0.112

The table shows summary statistics of the firm-level characteristics by country for the whole sample. Leverage is calculated as the ratio of total debt to book assets. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant and equipment. *Tangibility* is calculated as the ratio of total property, plant and equipment to book assets while *Profitability* is defined as the ratio of EBITDA to total book assets.

Table 3: Country-level Summary Statistics

Country	t	Summary Statistic	Stock Market Turnover Ratio	Stock Market Capitalization	Bank z -score	Political Stability	Strength of Legal Rights	Control of Corruption	FDIRR	Corporate Tax Rate
Argentina	10	Mean:	10.74	32.78	4.05	-0.19	4	-0.45	0.03	35
		Standard Deviation:	8.95	22.61	1.55	0.32	0	0.05	0.003	0
Australia	10	Mean:	83.69	115.20	10.84	0.92	9.01	2.01	0.15	30
		Standard Deviation:	11.22	16.15	2.71	0.08	0.29	0.10	0.001	0
Brazil	10	Mean:	50.82	52.86	22.73	-0.11	3	-0.02	0.09	34
		Standard Deviation:	16.99	15.76	1.50	0.22	0	0.11	0.006	0
Canada	10	Mean:	74.94	112.92	20.38	0.98	7	1.99	0.20	34.54
		Standard Deviation:	16.27	17.83	2.25	0.11	0	0.09	0.006	3.08
Egypt	10	Mean:	37.70	52.26	35.61	-0.75	3	-0.55	0.08	20
		Standard Deviation:	17.89	23.86	3.35	0.28	0	0.13	0.01	0
Indonesia	10	Mean:	53.12	29.33	2.52	-1.32	5	-0.80	0.33	28.8
		Standard Deviation:	16.49	9.91	0.31	0.47	0	0.17	0.01	2.10
India	10	Mean:	102.74	65.15	37.34	-1.21	7.05	-0.44	0.32	34.78
		Standard Deviation:	31.73	26.62	3.42	0.15	0.68	0.08	0.02	1.31
Mexico	10	Mean:	27.64	27.80	25.66	-0.52	5.04	-0.27	0.22	30.5
		Standard Deviation:	3.63	8.57	4.67	0.27	0.38	0.07	0.006	2.59
Malaysia	10	Mean:	32.32	131.52	28.70	0.24	10	0.20	0.31	26.8
		Standard Deviation:	7.98	13.29	2.06	0.20	0	0.15	0.04	1.39
New Zealand	9	Mean:	43.64	37.72	20.60	1.22	10	2.36	0.20	31.6
		Standard Deviation:	18.65	6.99	8.05	0.12	0	0.07	0.03	1.89
Peru	10	Mean:	6.65	46.31	12.75	-0.96	5.42	-0.26	0.092	31.2
		Standard Deviation:	1.72	19.05	0.87	0.16	1.17	0.08	0	2.78
U.S.A	10	Mean:	208.89	121.14	24.82	0.29	9	1.50	0.08	35
		Standard Deviation:	91.84	14.32	1.73	0.28	0	0.28	0.003	6.67
South Africa	10	Mean:	51.10	196.59	11.11	-0.07	7	0.29	0.07	34.78
		Standard Deviation:	9.53	42.46	6.18	0.17	0	0.18	0.001	3.12

The table shows summary statistics of the country-level characteristics for the whole sample. t represents the number of years for which the country-specific data is available. *Stock Market Turnover Ratio* is defined as the total value of shares traded during a given period divided by the average market capitalization for that period. *Stock Market Capitalization* is defined as the total value of all listed shares in a stock market as a percentage of GDP. *Bank z -score* captures the probability of default of a country's commercial banking system. A higher z -score implies a lower probability of insolvency and hence a more stable commercial banking system. *Political Stability* captures perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders and thus facilitate lending. *Control of Corruption* captures the strength and the effectiveness of a country's policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* is the corporate tax rate for each country.

Table 4: Firm-level Summary Statistics (By Quartile)

Variable	Summary Statistic	1 st Quartile	2 nd Quartile	3 rd Quartile	4 th Quartile
Firm Size	Mean:	0.622	1.408	1.917	2.716
	Standard Deviation:	0.991	0.531	0.506	0.636
<u>Total Debt</u> Assets	Mean:	0.283	0.197	0.211	0.278
	Standard Deviation:	0.383	0.234	0.231	0.231
Investment	Mean:	9.780	13.073	22.196	188.198
	Standard Deviation:	142.458	259.496	120.107	783.282
Tangibility	Mean:	0.252	0.293	0.312	0.340
	Standard Deviation:	0.253	0.250	0.248	0.243
Profitability	Mean:	-0.384	-0.060	0.050	0.116
	Standard Deviation:	0.686	0.336	0.194	0.104
IFRS Compliant		2,552	2,570	2,419	2,634
U.S. GAAP Compliant		8,631	9,471	10,031	11,839
N		25,116	26,146	26,514	28,406

The table shows the summary statistics for the firm-level characteristics for each quartile. Leverage is calculated as the ratio of total debt to book assets. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant and equipment. *Tangibility* is calculated as the ratio of total property, plant and equipment to book assets while *Profitability* is defined as the ratio of EBITDA to total book assets.

Table 5: OLS Regressions (Full Sample): Total Debt to Assets - Financial Development Proxies

Variable	(1)	(2)	(3)	(4)	(5)
Stock market turnover ratio	0.00007 (0.09)			-0.001*** (-4.43)	
Stock market capitalization		0.00005 (0.85)			-0.001*** (-4.44)
Bank <i>z</i>-score			0.0006 (1.61)	0.002*** (6.04)	0.002*** (6.22)
Firm Size	0.038*** (13.91)	0.038*** (13.89)	0.043*** (15.06)	0.043*** (15.05)	0.043*** (15.08)
Investment	-0.00009 (-1.90)	-0.00009 (-1.95)	0.00002 (0.44)	0.00002 (0.39)	0.00002 (0.47)
IFRS	-0.018 (-1.30)	-0.001 (-0.09)	-0.011 (-0.80)	-0.013 (-0.90)	-0.007 (-0.52)
GAAP	0.100*** (5.92)	0.102*** (8.26)	0.075*** (5.83)	0.082*** (4.80)	0.077*** (6.01)
Tangibility	0.205*** (23.96)	0.205*** (24.00)	0.203*** (23.74)	0.203*** (23.70)	0.203*** (23.76)
Profitability	-0.241*** (-42.05)	-0.241*** (42.02)	-0.248*** (-42.50)	-0.248*** (-42.47)	-0.248*** (-42.48)
Political Stability	-0.018* (-2.01)	-0.031*** (-3.29)	-0.021* (-2.41)	-0.019* (-2.17)	-0.034*** (-3.72)
Strength of Legal Rights	0.007** (3.14)	0.011*** (4.37)	0.006** (2.94)	0.007** (2.88)	0.011*** (4.31)
Control of Corruption	0.066*** (9.35)	0.074*** (10.53)	0.058*** (8.45)	0.056*** (7.76)	0.066*** (9.22)
FDIRR	-0.009 (-0.16)	-0.036 (-0.62)	-0.166** (-2.64)	0.164** (2.61)	0.200** (3.16)
Corporate Tax Rate	0.005*** (6.20)	0.006*** (6.94)	0.004*** (3.82)	0.004*** (3.75)	0.004*** (4.62)
Firm Fixed-effects	Yes	Yes	Yes	Yes	Yes
Year Fixed-effects	Yes	Yes	Yes	Yes	Yes
N	106,182	106,182	106,182	106,182	106,182
<i>F</i> -statistic	-	-	-	18.23	25.87
<i>R</i> ²	20.24%	20.30%	20.41%	20.42%	20.49%

This table shows the results of the OLS regressions of *Leverage* measured as the ratio of total debt to assets on various combinations of the proxies for financial development – stock market turnover ratio, stock market capitalization as well as bank *z*-score – together with other firm- and country-level controls. *Stock Market Turnover Ratio* is ratio of the total value of shares traded during a given period to the average market capitalization for that period. *Stock Market Capitalization* is defined as the total value of all listed shares in a stock market as a percentage of GDP. *Bank *z*-score* captures the probability of default of a country's commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets. *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* captures the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders. *Control of Corruption* captures the strength and the effectiveness of a country's policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* is the corporate tax rate for each country. *t*-statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.

Table 6: OLS Regression (Quartiles): Total Debt to Assets - Stock Market Turnover Ratio

Variable	1 st Quartile	2 nd Quartile	3 rd Quartile	4 th Quartile
Stock market turnover ratio	-0.003*** (-5.45)	-0.001*** (-6.56)	-0.001*** (-4.68)	-0.001** (-3.00)
Bank z-score	0.001* (2.20)	0.001*** (5.47)	0.003*** (3.81)	0.002* (2.36)
Firm Size	0.086*** (5.69)	0.047*** (6.48)	0.041*** (3.39)	0.049*** (5.38)
Investment	0.0006 (0.66)	0.0004 (1.37)	0.00006 (0.32)	-0.0002 (-1.62)
IFRS	-0.015 (-0.45)	-0.002 (-0.06)	-0.003 (-0.12)	-0.059 (-1.87)
GAAP	0.783*** (8.32)	0.231*** (3.38)	0.074 (1.49)	0.151*** (3.55)
Tangibility	0.161*** (7.24)	0.165*** (10.80)	0.078*** (10.37)	0.162*** (11.26)
Profitability	-0.224*** (-21.39)	-0.241*** (-32.17)	-0.198*** (-34.90)	-0.155*** (-14.54)
Political Stability	-0.044*** (-5.99)	-0.037*** (-4.56)	-0.049** (-2.66)	-0.089*** (-4.63)
Strength of Legal Rights	0.030*** (4.99)	0.031*** (9.68)	0.011*** (5.59)	0.006* (2.33)
Control of Corruption	0.131*** (6.76)	0.182*** (7.90)	0.120*** (3.61)	0.017* (2.04)
FDIRR	0.769*** (4.21)	0.394** (2.74)	0.145*** (3.36)	0.411* (2.66)
Corporate Tax Rate	0.005* (2.28)	0.004** (2.61)	0.002** (2.44)	0.003*** (3.62)
Firm Fixed-effects	Yes	Yes	Yes	Yes
Year Fixed-effects	Yes	Yes	Yes	Yes
N	25,116	26,146	26,514	28,406
F-statistic	18.87	37.60	27.42	20.32
R ²	19.15%	13.39%	17.06%	13.41%

The table shows the results of the regression of various firm- and country-specific characteristics on the ratio of total debt to assets for each of the four quartiles. *Stock Market Turnover Ratio* is defined as the total value of shares traded during a given period divided by the average market capitalization for that period. *Bank z – score* captures the probability of default of a country’s commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets while *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* captures perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders and thus facilitate lending. *Control of Corruption* captures the strength and the effectiveness of a country’s policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* is the corporate tax rate for each country. *t*-statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.

Table 7: OLS Regression (Quartiles): Total Debt to Assets - Stock Market Capitalization

Variable	1 st Quartile	2 nd Quartile	3 rd Quartile	4 th Quartile
Stock market capitalization	-0.002*** (-5.48)	-0.002*** (-6.51)	-0.002*** (-8.02)	-0.001** (-3.01)
Bank z-score	0.001*** (5.80)	0.001*** (4.00)	0.003*** (4.15)	0.002** (2.87)
Firm Size	0.046*** (3.96)	0.013*** (4.06)	0.051*** (4.25)	0.094*** (9.28)
Investment	0.0007 (0.77)	0.0002 (1.75)	0.0001 (0.85)	-0.0001 (-1.58)
IFRS	-0.160*** (-4.24)	-0.028*** (-3.36)	-0.042 (-1.49)	-0.019 (-0.57)
GAAP	0.326*** (9.93)	0.077*** (9.76)	0.113*** (11.67)	0.095*** (3.38)
Tangibility	0.180*** (8.20)	0.186*** (17.95)	0.216*** (14.78)	0.225*** (13.97)
Profitability	-0.229*** (-22.41)	-0.247*** (-33.00)	0.155*** (-14.54)	-0.225*** (-24.88)
Political Stability	-0.168*** (-6.95)	-0.111*** (-9.86)	-0.039*** (-4.22)	-0.046*** (-6.57)
Strength of Legal Rights	0.063*** (10.38)	0.041*** (10.25)	0.038*** (9.39)	0.068*** (9.44)
Control of Corruption	0.215*** (10.88)	0.231*** (10.08)	0.126*** (10.80)	0.188*** (11.65)
FDIRR	0.215 (1.34)	0.142** (2.28)	0.191 (1.45)	0.421** (2.67)
Corporate Tax Rate	0.007*** (7.95)	0.008*** (9.43)	0.008*** (4.37)	0.005*** (4.58)
Firm Fixed-effects	Yes	Yes	Yes	Yes
Year Fixed-effects	Yes	Yes	Yes	Yes
N	25,116	26,146	26,514	28,406
F-statistic	15.97	37.62	31.48	22.06
R ²	19.08%	14.19%	17.35%	13.51%

The table shows the results of the regression of various firm- and country-specific characteristics on the ratio of total debt to assets for each of the four quartiles. *Stock Market Capitalization* is defined as the total value of all listed shares in a stock market as a percentage of GDP. *Bank z - score* captures the probability of default of a country's commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets while *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* captures perceptions of the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders and thus facilitate lending. *Control of Corruption* captures the strength and the effectiveness of a country's policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* is the corporate tax rate for each country. t -statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.

Table 8: OLS Regression (Full Sample): Long-term Debt to Assets - Financial Development Proxies

Variable	(4)	(5)
Stock market turnover ratio	-0.002*** (-7.17)	
Stock market capitalization		-0.002*** (-7.19)
Bank <i>z</i>-score	0.003*** (5.26)	0.003*** (4.75)
Firm Size	0.036*** (7.64)	0.038*** (7.99)
Investment	0.000005 (0.06)	0.0001 (1.29)
IFRS	-0.109 (-4.39)	0.055* (2.05)
GAAP	0.448*** (10.43)	0.192*** (9.02)
Tangibility	0.116*** (8.22)	0.125*** (8.85)
Profitability	-0.661*** (-68.52)	-0.663*** (-68.95)
Political Stability	-0.066*** (-4.06)	-0.192*** (-11.11)
Strength of Legal Rights	0.012** (2.94)	0.015** (3.07)
Control of Corruption	0.139*** (10.04)	0.238*** (18.79)
FDIRR	0.437*** (4.20)	0.691*** (6.50)
Corporate Tax Rate	0.009*** (6.11)	0.010*** (9.71)
Firm Fixed-effects	Yes	Yes
Year Fixed-effects	Yes	Yes
N	106,182	106,182
<i>F</i> -statistic	13.60	16.93
<i>R</i> ²	39.27%	39.63%

The table shows the results of the regression of various firm- and country-specific characteristics on the ratio of long-term debt to book assets for the full sample. *Stock Market Turnover Ratio* is ratio of the total value of shares traded during a given period to the average market capitalization for that period. *Stock Market Capitalization* is defined as the total value of all listed shares in a stock market as a percentage of GDP. *Bank *z* – score* captures the probability of default of a country’s commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets. *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* captures the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders. *Control of Corruption* captures the strength and the effectiveness of a country’s policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* is the corporate tax rate for each country. *t*-statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.

Table 9: OLS Regression (Quartiles): Long-term Debt to Assets - Stock Market Turnover Ratio

Variable	1 st Quartile	2 nd Quartile	3 rd Quartile	4 th Quartile
Stock market turnover ratio	-0.002* (-2.09)	-0.002* (-2.48)	-0.001** (-3.21)	-0.001** (-3.21)
Bank z-score	0.001*** (3.01)	0.002*** (5.82)	0.003*** (5.36)	0.002** (2.67)
Firm Size	0.024** (3.13)	0.028*** (3.71)	0.052*** (5.95)	0.053*** (7.27)
Investment	0.0002 (0.43)	0.0002 (0.36)	0.0002 (1.22)	0.00007 (1.15)
IFRS	-0.006 (-0.32)	-0.007 (-0.50)	-0.015 (-0.79)	-0.041* (-1.96)
GAAP	0.116* (2.48)	0.172* (2.46)	0.018 (0.50)	0.081* (2.39)
Tangibility	0.082*** (7.38)	0.166*** (10.64)	0.166*** (15.69)	0.157*** (13.56)
Profitability	-0.033*** (-6.40)	-0.029*** (-10.01)	-0.073*** (4.64)	0.030*** (-4.27)
Political Stability	-0.103* (-2.35)	-0.094*** (-3.28)	-0.058*** (-4.32)	-0.037* (-2.38)
Strength of Legal Rights	0.006*** (4.01)	0.004*** (3.46)	0.006*** (4.43)	0.006* (2.33)
Control of Corruption	0.012** (3.16)	0.014*** (5.06)	0.045*** (4.26)	0.051*** (3.96)
FDIRR	0.081* (2.23)	0.202*** (3.95)	0.349*** (3.76)	0.139* (2.50)
Corporate Tax Rate	0.003* (2.08)	0.003*** (3.32)	0.002** (2.79)	0.004*** (4.78)
Firm Fixed-effects	Yes	Yes	Yes	Yes
Year Fixed-effects	Yes	Yes	Yes	Yes
N	25,116	26,146	26,514	28,406
F-statistic	12.34	24.74	59.50	25.36
R ²	13.74%	13.99%	17.13%	17.25%

The table shows the results of the regression of various firm- and country-specific characteristics on the ratio of long-term debt to book assets for the full sample. *Stock Market Turnover Ratio* is ratio of the total value of shares traded during a given period to the average market capitalization for that period. *Bank z – score* captures the probability of default of a country’s commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets. *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* is defined to capture the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders. *Control of Corruption* captures the strength and the effectiveness of a country’s policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* captures the corporate tax rate for each country. *t*-statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.

Table 10: OLS Regression (Quartiles): Long-term Debt to Assets - Stock Market Capitalization

Variable	1 st Quartile	2 nd Quartile	3 rd Quartile	4 th Quartile
Stock market capitalization	-0.002* (-2.11)	-0.002* (-2.50)	-0.002*** (-3.96)	-0.001** (-2.76)
Bank <i>z</i>-score	0.002*** (3.52)	0.002*** (5.90)	0.003*** (5.62)	0.002** (2.68)
Firm Size	0.028*** (4.90)	0.028*** (4.76)	0.057*** (6.67)	0.054*** (7.34)
Investment	0.0002 (0.45)	0.0003 (0.67)	0.0002 (1.68)	0.00005 (0.97)
IFRS	-0.011 (-0.58)	-0.009 (-0.55)	-0.009 (-0.49)	-0.027 (-1.17)
GAAP	0.064*** (3.91)	0.048 (1.91)	0.029 (1.61)	0.063** (3.14)
Tangibility	0.084*** (7.66)	0.121*** (13.55)	0.168*** (15.93)	0.158*** (13.65)
Profitability	-0.034*** (-6.64)	-0.027*** (-8.78)	-0.051*** (-9.39)	-0.032*** (-7.09)
Political Stability	-0.090*** (-5.40)	-0.039*** (-8.56)	-0.035** (-2.60)	-0.025** (-2.40)
Strength of Legal Rights	0.018*** (4.12)	0.005* (2.01)	0.022*** (4.91)	0.032*** (3.35)
Control of Corruption	0.036** (2.58)	0.062*** (4.36)	0.031*** (3.21)	0.017*** (3.22)
FDIRR	0.398*** (3.55)	0.212** (2.93)	0.420*** (4.42)	0.642*** (3.73)
Corporate Tax Rate	0.005** (3.24)	0.003** (3.24)	0.003** (2.45)	0.007** (2.88)
Firm Fixed-effects	Yes	Yes	Yes	Yes
Year Fixed-effects	Yes	Yes	Yes	Yes
N	25,116	26,146	26,514	28,406
F-statistic	13.95	23.81	43.74	33.15
R ²	14.91%	14.08%	17.75%	17.44%

The table shows the results of the regression of various firm- and country-specific characteristics on the ratio of long-term debt to book assets for the full sample. *Stock Market Capitalization* is defined as the total value of all listed shares in a stock market as a percentage of GDP. *Bank *z* - score* captures the probability of default of a country's commercial banking system. *Firm size* is calculated as the log of book assets. *Investment* is calculated as the sum of capital expenditures, increase in investments and acquisitions less sale of property, plant, equipment. *IFRS* is an indicator for firms that follow IFRS compliant accounting standards while *GAAP* is an indicator for firms that follow U.S. GAAP compliant accounting standards. *Tangibility* is the ratio of total property, plant and equipment to book assets. *Profitability* is the ratio of EBITDA to total book assets. *Political Stability* is defined to capture the likelihood that the government will be destabilized or overthrown by unconstitutional or violent means. *Strength of Legal Rights* measures the degree to which collateral and bankruptcy laws protect the rights of borrowers and lenders. *Control of Corruption* captures the strength and the effectiveness of a country's policy and institutional framework to prevent and combat corruption. *FDIRR* is the foreign direct investment regulatory restrictiveness index which measures statutory restrictions on foreign direct investment. *Corporate Tax Rate* captures the corporate tax rate for each country. *t*-statistics are reported in parentheses. *, ** and *** indicate significance at the 5%, 1% and 0.1% levels respectively.